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Withania somnifera and Withaferin A: Biomarkers, Cortisol Dynamics, and Neuroprotection

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ABSTRACT

Withania somnifera, popularly designated as Ashwagandha or Indian Ginseng, holds central status as a premier Rasayana in traditional Ayurvedic jurisprudence. This research clarifies the clinical and molecular basis of its adaptogenic and neuroprotective properties.

AYURVEDIC TRADITION AND ETHNOBOTANY

Withania somnifera has been traditionally used for over 3,000 years in Ayurveda as a rejuvenating rasayana to promote longevity, enhance cognitive vigor, and relieve nervous exhaustion. Indigenous practitioners prepare fine root powder mixed with clarified butter (ghee) or warm milk for restorative convalescence.

IN VITRO CELLULAR MODELS

Phytochemical fractions containing withaferin A demonstrated significant neuroprotection against beta-amyloid-induced toxicity in human neuroblastoma SH-SY5Y cell lines. In vitro assays demonstrated that withaferin A suppressed microglial nitric oxide synthase and reduced pro-inflammatory cytokines.

HUMAN CLINICAL TRIAL FINDINGS

In a prospective, randomized, double-blind, placebo-controlled clinical trial evaluating 64 healthy adult subjects with a history of chronic mental stress, administration of 300 mg of high-concentration Withania somnifera root extract twice daily for 60 days resulted in a 27.9% reduction in serum cortisol levels compared to the placebo group. Participants reported substantial improvements on standardized stress-assessment scales.

IN VIVO STRESS ASSAYS

In Sprague-Dawley rodents subjected to chronic restraint stress, Withania somnifera administration attenuated lipid peroxidation in the hippocampus and prevented stress-induced adrenal hypertrophy.

CONCLUSION

The integration of traditional Ayurvedic claims with verified randomized human clinical evidence substantiates Withania somnifera as an authentic adaptogen driven by steroidal withanolides.